



Oral Surgery

Surgical anatomy in relation to local anesthesia

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Surgical anatomy in relation to local anesthesia

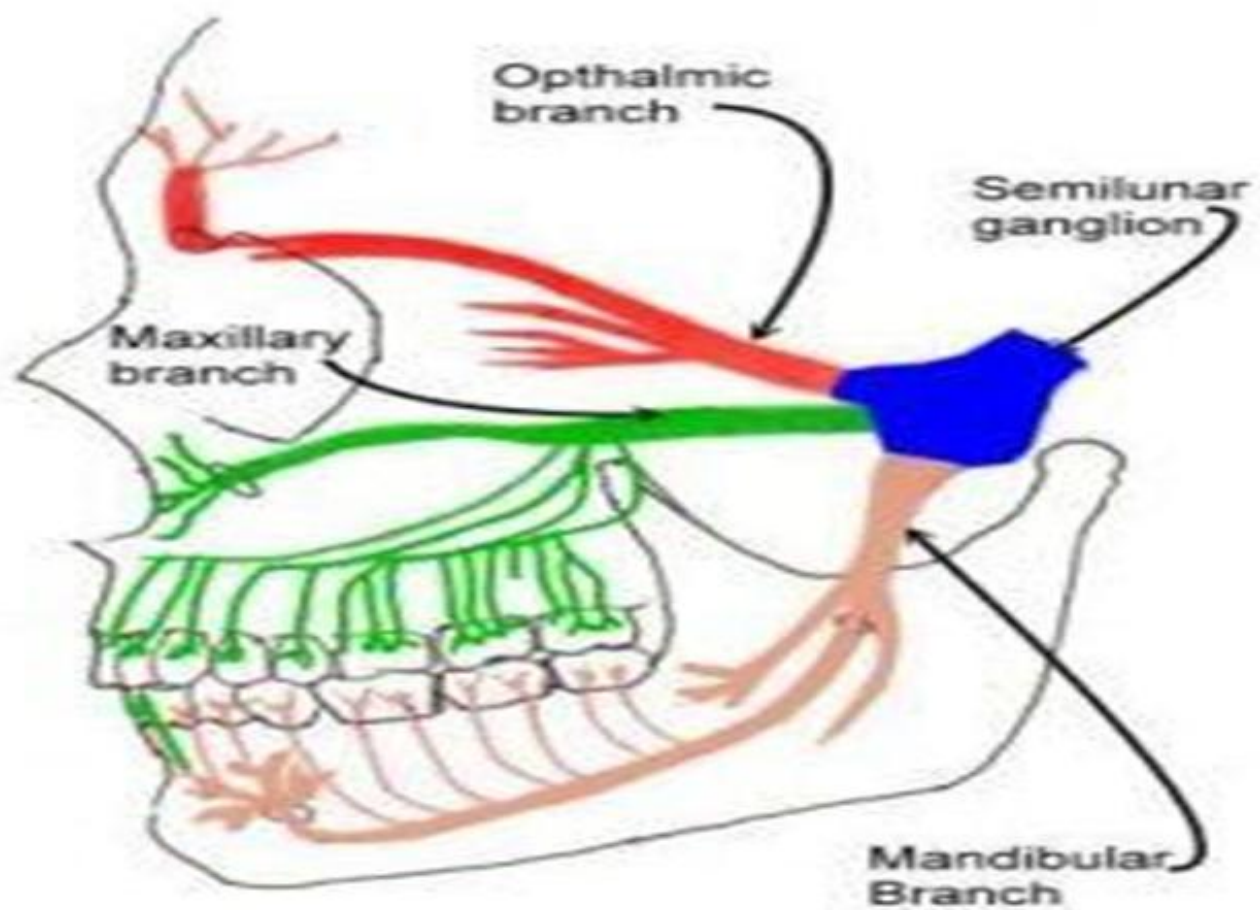


- The management of pain through local anesthesia by dental professionals requires a thorough knowledge of the anatomy of the skull, trigeminal nerve and related tissues.
- The skull bones involved in the local anesthetic administration are the maxilla, palatine, and the mandible, soft tissues of the face and oral cavity may serve the dental professional as initial landmarks to visualize and palpate for local anesthesia.

- The dentist must know the location of certain adjacent soft tissue structures such as a major blood vessel and glandular tissue and also should have an increased level of knowledge of the nervous system of the head and neck especially the **fifth (V) cranial or trigeminal nerve** which is the largest cranial nerve composed of a large sensory root and a small motor root(mixed), which supplies sensation to meninges, skin of the anterior part of the head, nasal and oral cavities and the teeth. It provides motor innervations to TMJ, the muscles of mastication and other muscles in the region.

The trigeminal nerve divided into three branches which are:-

- 1- Ophthalmic nerve
- 2- Maxillary nerve
- 3- Mandibular nerve



Supraorbital
foramen

Optthalmic branch (V_1)
through superior orbital fissure

Maxillary branch
(V_2) through
foramen rotundum

Infraorbital
foramen

Mandibular branch
(V_3) through
foramen ovale

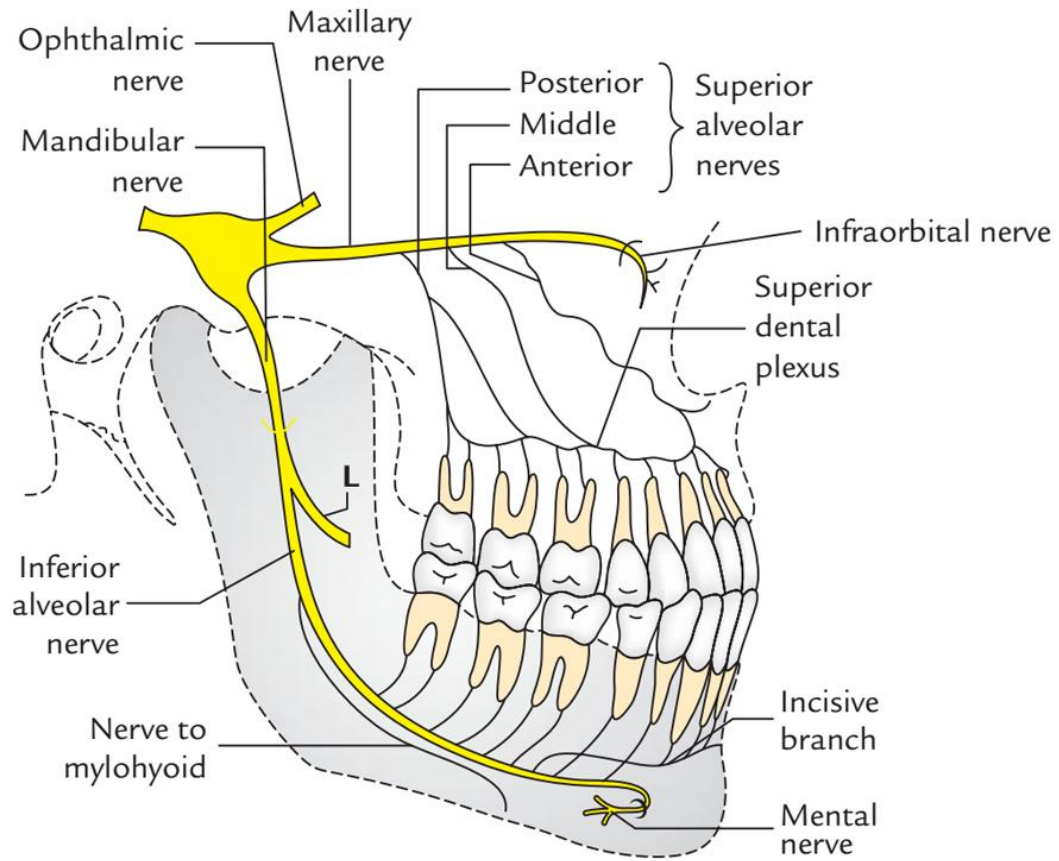
Mental foramen

1-Ophthalmic nerve:

- It is the smallest branch of the trigeminal nerve, leave the skull through the superior orbital fissure, it transmit sensory innervations from eyeballs, conjunctiva, skin of the upper face and anterior scalp, the lining of the upper part of the nasal cavity and paranasal air sinuses, and the meninges of the anterior cranial fossa.
- The ophthalmic nerve have the following terminal branches
 - Nasociliary nerve
 - Lacrimal nerve
 - Frontal nerve

2- Maxillary nerve

- It is the second division of the trigeminal nerve and it leaves the skull through the foramen rotundum, is purely sensory and is intermediate in size between ophthalmic and mandibular branches.
- It transmit sensory fibers from the skin of the face between the lower eyelid and the mouth, from the nasal cavity and the sinuses, **from maxillary teeth and maxilla.**



- The branches of the maxillary nerve subdivide into groups according to their **location of origin**, as follows:

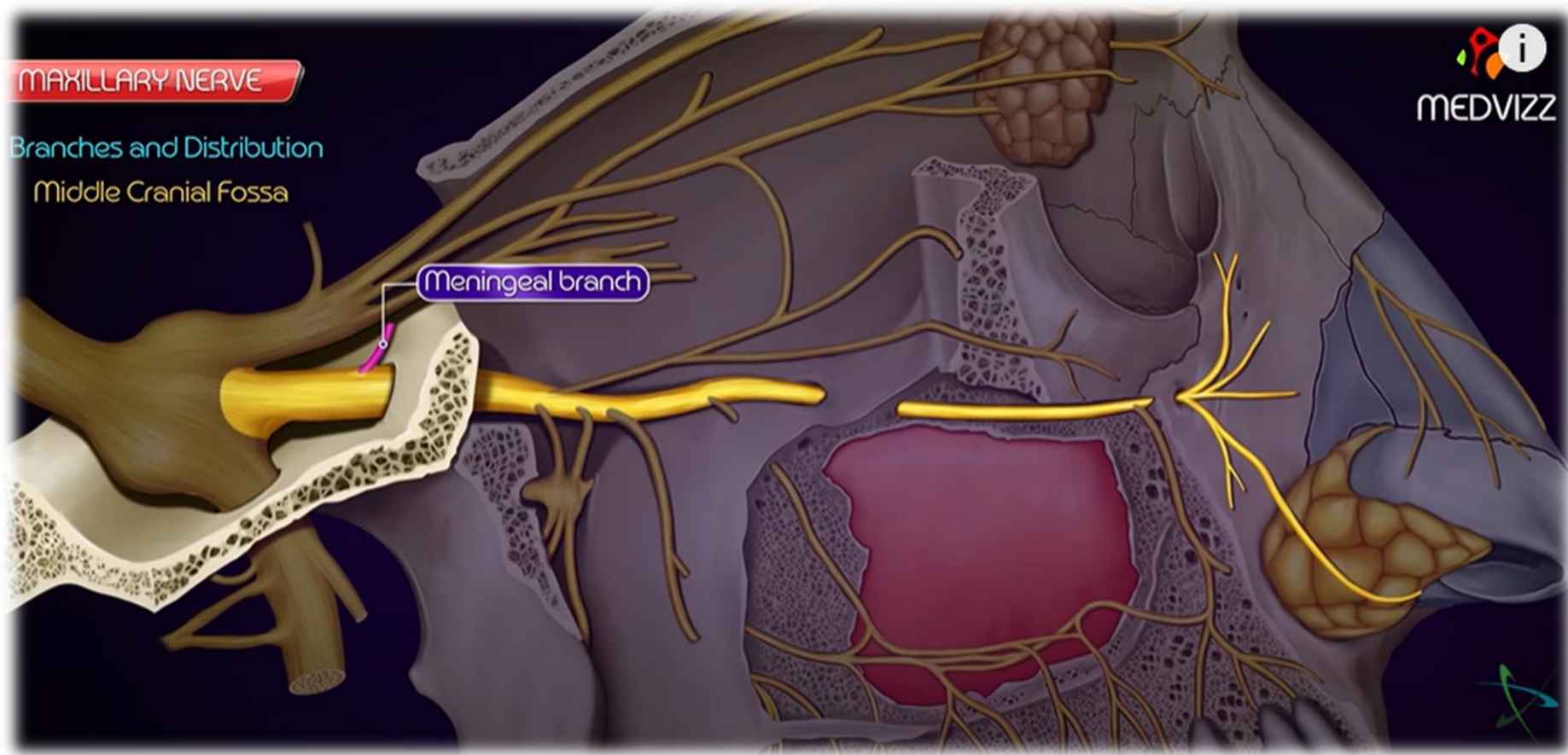
1) Cranium:

- While coursing through the middle cranial fossa, the maxillary nerve gives rise to its smallest branch, **the middle meningeal nerve** near foramen rotundum. This branch supplies the dura mater of the middle cranial fossa.

MAXILLARY NERVE

Branches and Distribution
Middle Cranial Fossa

Meningeal branch



2) Pterygopalatine fossa:

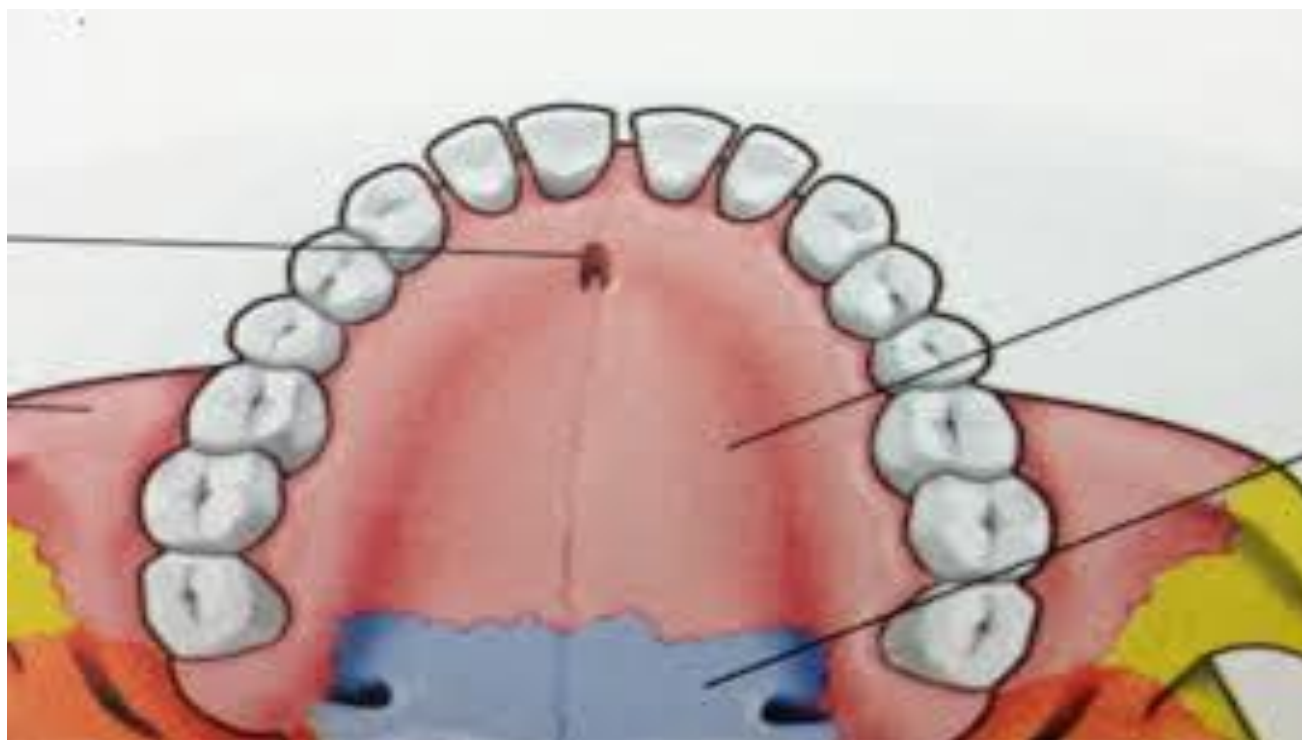
- The pterygopalatine fossa holds the pterygopalatine ganglion, the branches are:

- **a-Orbital Branches:**

- These small branches travel through the inferior orbital fissure and contribute to the innervation of orbital wall, sphenoidal sinus, and ethmoidal sinus.

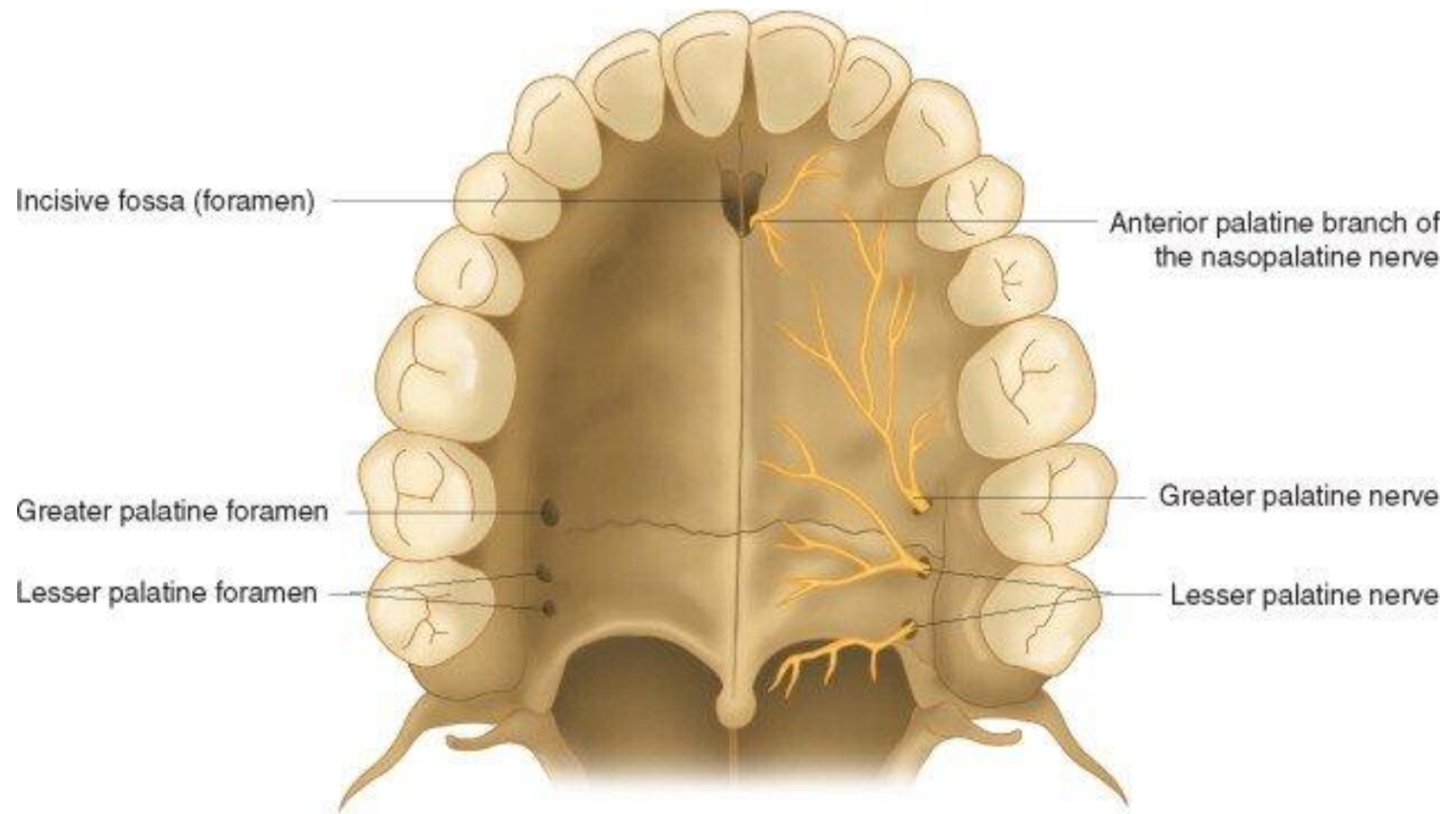
- **b-Nasal branches:**
- They include medial and lateral posterior superior nasal nerves and nasopalatine nerve. The nasopalatine nerve supplies the roof of the nasal cavity, nasal septum and the floor of the nasal cavity then it enters the incisive canal through which it passes into the oral cavity via the incisive foramen, located in the midline of hard palate about 1 cm posterior to maxillary central incisors.

- The right and the left nasopalatine nerves emerge together through the foramen and provide sensory innervations to palatal mucosa in the region of a central incisor, lateral incisors, and canines.



- **c-Palatine nerves:**
- They are the greater palatine nerve and the lesser palatine nerve. The greater palatine nerve provides sensory supply to the hard palate, it emerges on hard palate through the greater palatine foramen which is usually located about 1 cm towards the palatal midline, just distal to second molar.

- The nerve supplies sensory innervations to palatal soft tissue and bone up to the canine where it communicates with terminal fibers of nasopalatine nerves it also provide sensory innervations to some parts of soft palate. It supplies all the hard palate except that in the incisor area. The lesser palatine nerve supplies sensory innervations to the soft palate tonsil and uvula.





Anatomic area innervated by greater palatine nerve

- **d- Posterior superior alveolar nerves**
- These nerves are 2 -3 in number they emerge through the pterygomaxillary fissure and reach the posterior surface of the maxilla and remains external to bone, to provide sensory innervations to and adjacent facial mucosal surfaces. The other buccal gingiva in maxillary molar region branch enters maxilla and provide sensory innervations to the alveolar bone, buccal periodontal ligament and pulpal tissue of maxillary molars, with the exception of mesiobuccal root of the first molar.



**Anatomic area innervated by posterior
superior alveolar nerves**

- e-Ganglionic Branches
- f- Zygomatic Branch
- g-Pharyngeal Nerve.

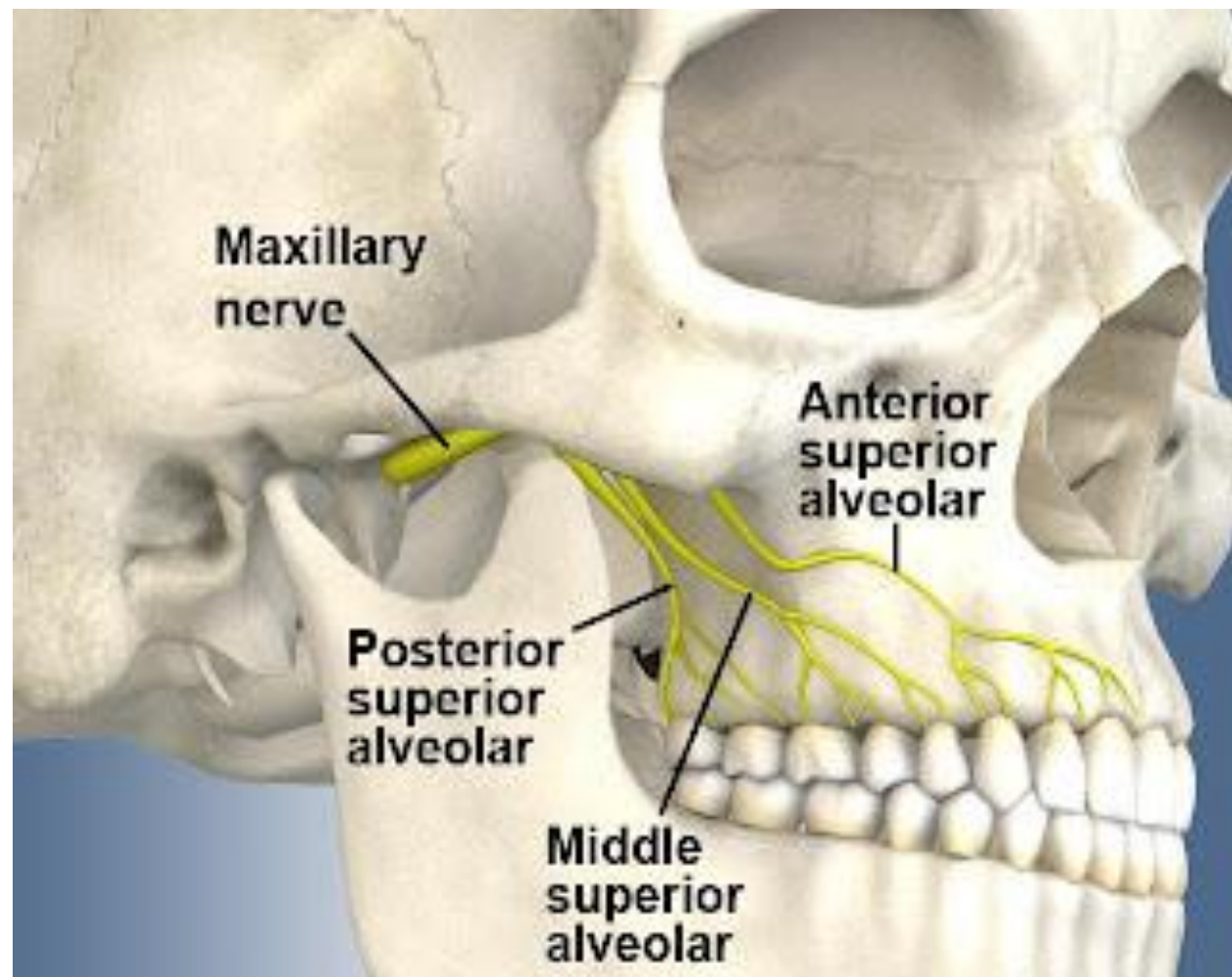
- 3) The floor of the orbit:
- After exiting the pterygopalatine fossa through the inferior orbital fissure, the maxillary nerve enters the orbit as infraorbital nerve which is its terminal branch. In the floor of orbit the infraorbital nerve gives off following two branches:

- a) Middle superior alveolar nerve: which originated in the infraorbital canal and run downward supplying adjacent lining of maxilla sinus, pulp of premolars and mesiobuccal root of the first molar, buccal periodontal tissues, buccal soft tissues and alveolar bone in the premolar region.



Anatomic area innervated by middle superior alveolar nerve

- b) Anterior superior alveolar nerve :-It a relatively larger branch and given off in the infraorbital canal, it provides innervations to the central and lateral incisors and canines, as well as sensory innervations to the adjacent lining of maxillary sinus, buccal periodontal tissues, buccal bone and buccal gingiva of these teeth.



Anesthesia of the Teeth and Supporting Tissues in the Maxillary Arch

Anesthetic Technique	Tooth and Supporting Tissues							
	Central Incisor	Lateral Incisor	Canine	First Premolar	Second Premolar	First Molar	Second Molar	Third Molar
Anterior Superior Alveolar	P X F X PL	P X F X PL	P X F X PL	P B PL	P B PL	P B PL	P B PL	P B PL
Middle Superior Alveolar	P F PL	P F PL	P F PL	P X B X PL	P X B X PL	P B PL	P B PL	P B PL
Posterior Superior Alveolar	P F PL	P F PL	P F PL	P B PL	P B PL	P X* B X PL	P X B X PL	P X B X PL
Greater Palatine	P F PL	P F PL	P F PL	P B PL X	P B PL X	P B PL X	P B PL X	P B PL X
Nasopalatine	P F PL X	P F PL X	P F PL X	P B PL	P B PL	P B PL	P B PL	P B PL

P: pulp, F: facial, PL: palatal

Anatomical variations

- In some cases, the infraorbital foramen, usually single, can be two to three in number. This variation is particularly significant during the infraorbital nerve block, which is often used to induce regional anesthesia of face.

- The greater palatine nerve can sometimes give off branches to innervate upper molar and premolar teeth. This variation is adequate for superior alveolar nerve block. Similarly, the nasopalatine nerve, in some cases, innervates incisor teeth; thus, a nasopalatine nerve block is essential to complete anesthesia of incisor teeth.

3- Mandibular nerve

- It is the largest of the three divisions of the trigeminal nerve and exits the cranial cavity through the foramen ovale.
- It is a **mixed** nerve and consist of two roots, a large sensory root and a small motor.

A-Buccal nerve

- Providing sensory innervation to the buccal gingiva of the mandibular molars and buccal vestibule.

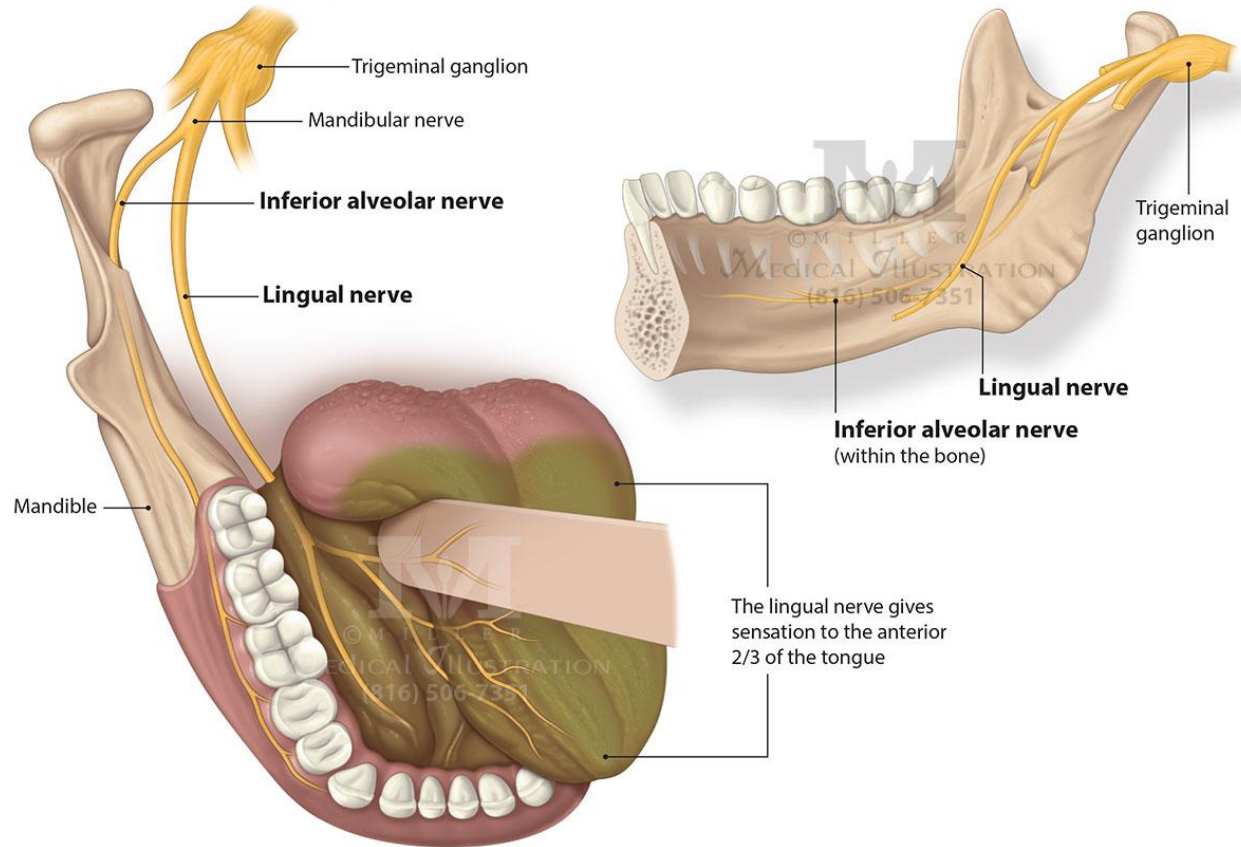
B-Lingual nerve

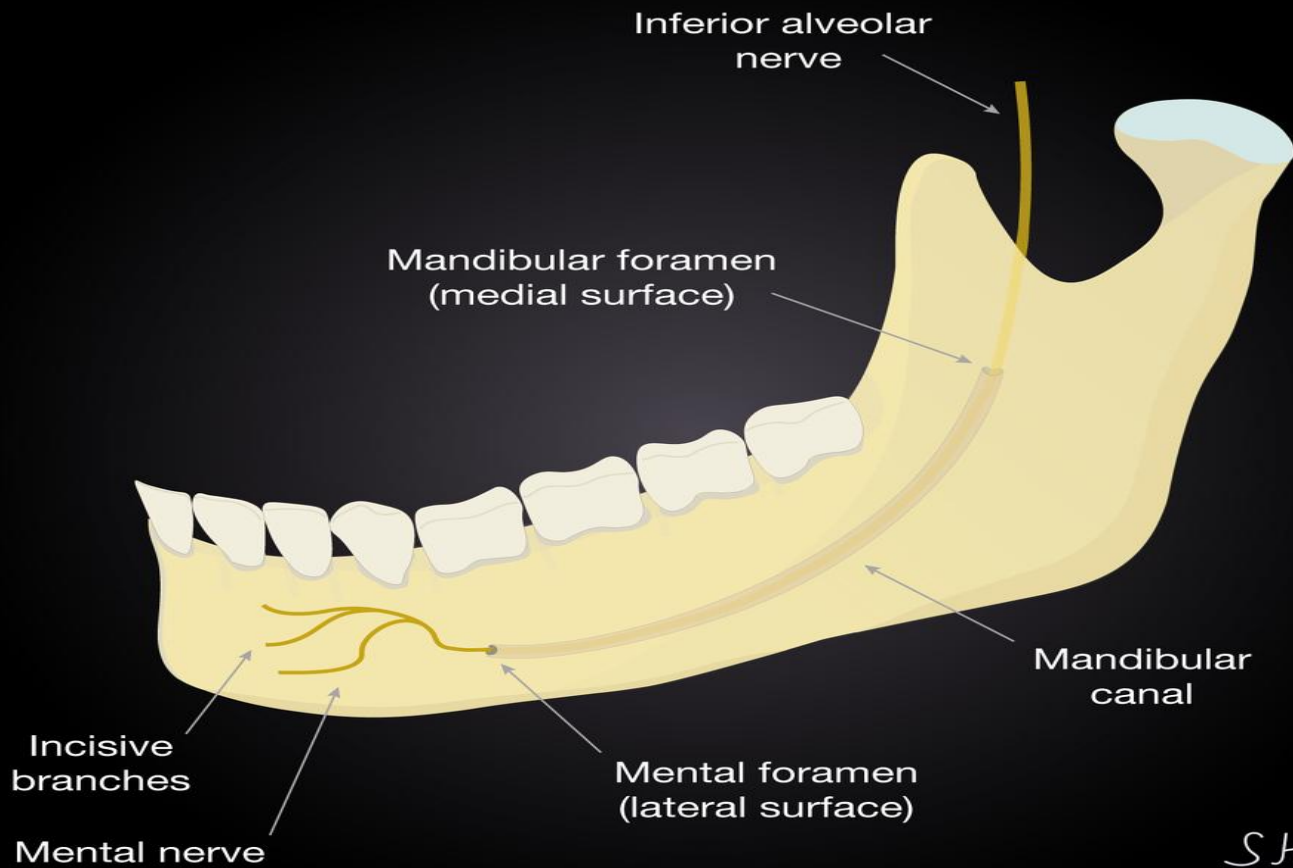
- Which provides sensory innervations to the gingiva on the lingual side of the mandible from the last molar to the central Incisor, the mucous membrane the floor of the mouth, to the sublingual and of submandibular salivary glands, and to the mucous membrane of the anterior 2/3rds of the tongue.

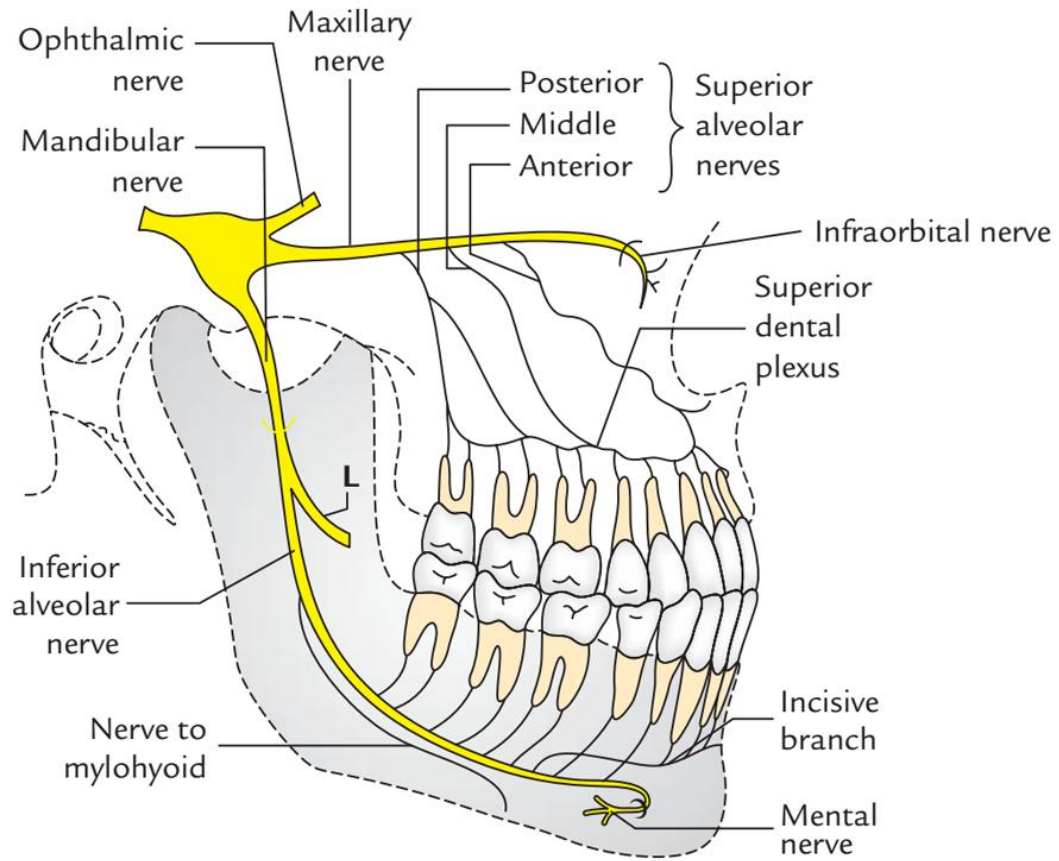
C-Inferior alveolar nerve

which is the largest branch of the mandibular nerve, it is accompanied by inferior alveolar artery and inferior alveolar vein in the mandibular canal and these three structures together are referred as inferior alveolar neurovascular bundle.

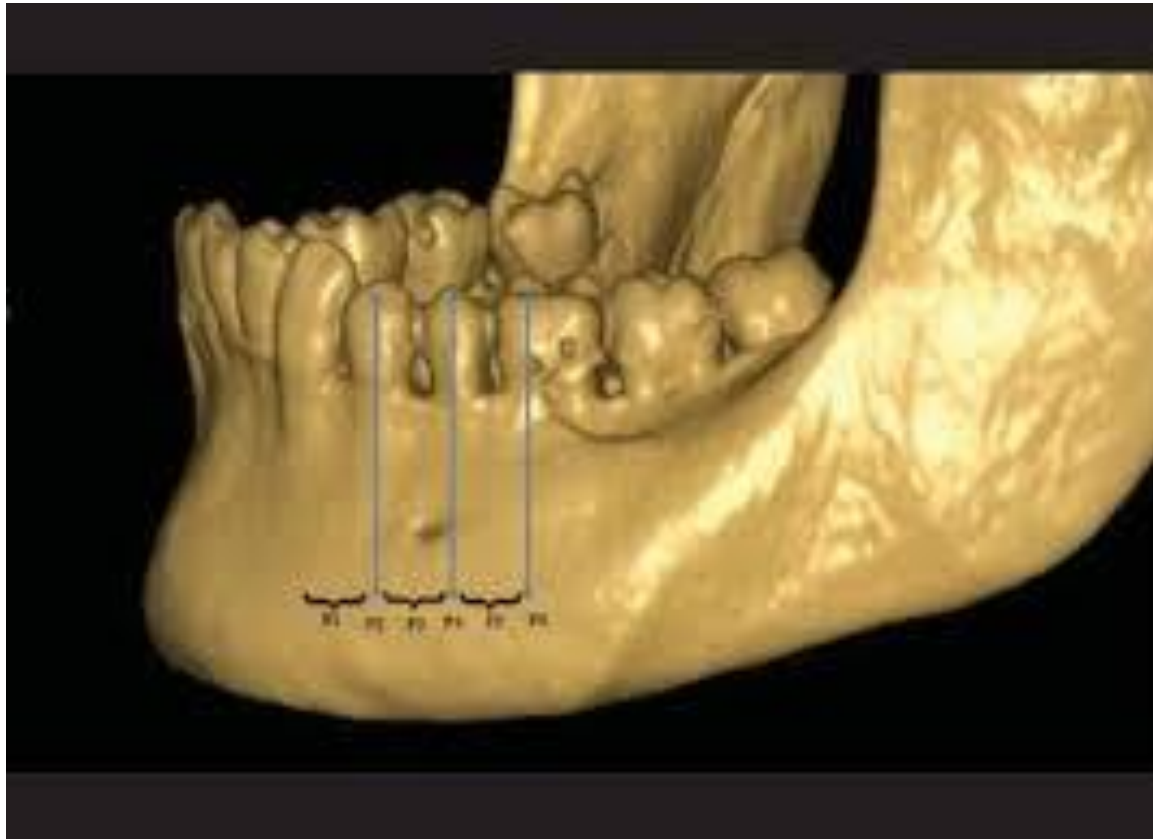
The Lingual and Inferior Alveolar Nerves







It supplies the inferior portion of the ramus of the mandible, entire body of the mandible, and pulps of the mandibular teeth, the nerve artery and vein travel in the mandibular canal as far forward as mental foramen which is located at a point below and between the roots of the premolars where the nerve divided into its terminal branches which are:



- 1 Mental nerve which supplies the skin of the chin, skin and the **mucous membrane of the lower lip, and the buccal mucosa from the incisor to the premolars.**
- 2 Incisive nerve which is the smaller terminal branch and the continuation of the inferior alveolar nerve within the substance of the body of the mandible anterior to the mental foramen it supplies the **pulps of the anterior teeth, central and lateral incisors and canine, periodontal ligament and the overlying soft tissue anterior to the mental foramen.**



Skin and mucous
membrane of the
lower lip

Skin of chin

